

Math 310: Differential Equations

Instructor: Benjamin Ridenhour

Fall 2024

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Office Hours: TR 4:00–5:00pm (or by appt.)

Office: Brink Hall 327

Web: <https://bit.ly/2scoPgS>

Class Hours: MWF 3:30–4:20am

Class Room: TLC 249

Course Description

Dynamical systems surround us in every aspect of our life. This first course on differential equations equips students to critically think about, model, and analyze simple systems of differential equations. The book used for this course highlights certain common, real-world dynamic systems which are of interest to many STEM students. Additionally, the book provides numerous computational examples which utilize software such as MATLAB or *Mathematica* to either solve or graph systems.

The structure of the class is as follows. Homework will be assigned for each class meeting. **All homework will be assigned and graded in the MyLab Math platform, thus it is critical for you to sign up for an account immediately.** I expect you to have read the appropriate section materials and done the homework *prior to* each class period. Class periods will be used for minimal lecturing, discussion of the homework set, and strategies for solving problems. You should be prepared to participate in discussions. Student assessment is based around completion of the homework, 3 exams, and a final (comprehensive) exam (see below for details). Dates of the 3 exams will be determined based on progress of the class and announced at least a week in advance (they will roughly occur after every 4 weeks).

Required Materials

- *Differential equations and boundary value problems: computing and modeling*. 6th edn. by CH Edwards, DE Penney, and DT Calvis. Pearson Education, Inc.: London.

NOTE: The textbook is available via inclusive access on Canvas

- **Subscription to Pearson MyLab Math** (*Note: Access to MyLab Math is free with purchase of a new book. Otherwise, the bookstore can sell access or you can pay for access directly via the Pearson website. See the directions for registering for the course at the end of the syllabus.*)

Prerequisites/Corequisites

Prerequisites: Math 175

Preferred: Math 275

Topics

1. First-order differential equations (Chapter 1, §1.1–1.6)
2. Mathematical models and numerical methods (Chapter 2, §2.1–2.6)
3. Linear equations of higher order (Chapter 3, §3.1–3.8)
4. Introduction to systems of differential equations (Chapter 4 §4.1–4.3)
5. Linear systems of differential equations (Chapter 5, §5.1–5.3)
6. Laplace transform methods (Chapter 7, §7.1–7.4)

Course Objectives/Learning Outcomes

Successful students will:

1. Learn how to model a dynamic physical phenomenon as a differential equation
2. Gain mastery of standard methods for solving initial value problems, both analytically and numerically.
3. Be able to analyze and interpret qualitative aspects of solutions to ODEs.

Final Exam Date

13 December (Friday), 3:00–5:00pm

Grading Policy

A typical grading scale will be used (i.e., $\geq 93\%$ = A, $(93,90\%]$ = A-, $(90,87\%]$ = B+, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 30% of your grade will be determined by completing homework.
- 45% of your grade will be determined by 3 equally-weighted exams.
- 25% of your grade will be determined by the final exam.